

- 1 Let a, b, c and d be integers satisfying $ad - bc > 1$. Show that at least one of a, b, c, d is not divisible by $ad - bc$. [2]

- 2 Let u_r be the number of ways to cover a rectangle of size $3 \times (2r)$ with $3r$ nonoverlapping rectangles of size 2×1 .

(i) Show that $u_1 = 3$ and $u_2 = 11$. [3]

(ii) Show that, for $r \geq 3$, $u_r = 3u_{r-1} + 2 + 2 \sum_{k=1}^{r-2} u_k$. [4]

(iii) Hence, show that $u_r = 4u_{r-1} - u_{r-2}$. [2]

- 3 Let n, k and r be positive integers, with $n \geq rk$. Let $S_r(n, k)$ denote the number of ways to put n distinguishable objects into k identical boxes such that there are at least r objects in each box.

(i) Show that $S_2(n, 2) = 2^{n-1} - n - 1$. [3]

(ii) Show that $S_2(n, 3) = \frac{3^{n-1} + n^2 + n + 1}{2} - (n+2)2^{n-2}$. [6]

(iii) If n is even, show that $S_2(n, \frac{1}{2}n) = 1 \times 3 \times 5 \times \dots \times (n-1)$. [3]

(iv) If n is an odd number such that $n \geq 3$, show that

$$S_2\left(n, \frac{n-1}{2}\right) = (1 \times 3 \times 5 \times \dots \times n) \frac{n-1}{6}. \quad [3]$$

- 4 (a) Use the substitution $x = \sin u$ to find $\int \frac{x}{(x^2 - 4)\sqrt{1 - x^2}} dx$. [3]

- (b) (i) It is given that

$$I_1 = \int_0^k \frac{2 + \sin kx}{4 + \sin kx + \cos kx} dx \text{ and } I_2 = \int_0^k \frac{2 + \cos kx}{4 + \sin kx + \cos kx} dx,$$

where k is a positive constant. By considering combinations of I_1 and I_2 , find an expression for I_1 in terms of k . [4]

(ii) Find the exact value of $\int_0^\pi \frac{3 + \cos x}{13 + 2 \sin x + 3 \cos x} dx$. [3]

- 5 (a) By differentiating a suitable series, obtain an expression for

$$1 + 2x + 3x^2 + \dots + nx^{n-1}$$

and find the sum of the series

$$1 - 2 + 3 - 4 + 5 - 6 + \dots + (-1)^{n-1} n$$

for the cases when n is odd and when n is even.

[4]

- (ii) Show that

$$1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + \dots + (-1)^{n-1} n^2 = A(-1)^{n+1} n(n+B),$$

where A and B are constants to be determined.

[4]

- (iii) Given that $1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$, find the sum of the series

$$1^3 - 2^3 + 3^3 - 4^3 + 5^3 - 6^3 + \dots + (-1)^{n-1} n^3$$

for the case when n is odd, giving your answer in factorised form.

[3]

- 6 A sequence of polynomials $p_0(x), p_1(x), p_2(x), \dots$ is defined by $p_0(x) = 1$, and

$$\frac{dp_n(x)}{dx} = np_{n-1}(x) \text{ and } \int_0^1 p_n(x) dx = 0 \text{ for all positive integers } n.$$

- (i) Prove by induction that $p_n(x+1) - p_n(x) = nx^{n-1}$ for all positive integers n .

[6]

It is given that $p_5(x) = x^5 - \frac{5}{2}x^4 + \frac{5}{3}x^3 - \frac{1}{6}x$.

- (ii) Obtain an expression for $p_6(x)$.

[2]

- (iii) Using the results in (i) and (ii), show that for any positive integer N ,

$$\sum_{r=1}^N r^5 = aN^6 + bN^5 + cN^4 + dN^2,$$

where a, b, c and d are constants to be determined.

[3]

- 7 A set S contains only distinct positive integers, where x of them are odd and y of them are even. It is known that the sum of all elements in the set is 654 321.

(i) Show that $x^2 + (y + 0.5)^2 \leq 654\,321.25$. [3]

It is given that $n = 21x + 24y$.

(ii) Using Cauchy-Schwarz inequality, show that n is bounded above by 25 782. [3]

(iii) For every pair of x and y that satisfies $96 \leq n \leq 114$, write down a possible set S . [3]

- 8 (i) Let a and b be integers such that $\gcd(a, b) = 1$. Prove that $\gcd(a^m, b^n) = 1$ for all positive integers m and n . [2]

- (ii) A *monic* polynomial is a polynomial whose leading coefficient (coefficient of the highest degree term) is 1.

Prove that any rational root of a monic polynomial with integer coefficients is an integer. [3]

- (iii) For any positive integer n and prime p , let $e_p(n)$ denote the exponent of p in the prime factorised form of n . For example, $e_2(40) = e_2(2^3 \times 5) = 3$. Also, let $\lfloor x \rfloor$ denote the largest integer less than or equal to x .

Explain why the series $\sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor$ is finite and show, using a counting argument or otherwise, that

$$e_p(n!) = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor.$$

Deduce the inequality $e_p(n!) < n$. [5]

- (iv) Hence, show that for any integer $n > 1$, the equation

$$\frac{x^n}{n!} + \frac{x^{n-1}}{(n-1)!} + \frac{x^{n-2}}{(n-2)!} + \cdots + \frac{x}{1!} + 1 = 0$$

has no rational roots. [5]

9 Let a, b, c, d, m be integers.

(i) Show that if $\gcd(a, m) = \gcd(b, m) = 1$, then $\gcd(ab, m) = 1$. [2]

(ii) Show that if $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $ac \equiv bd \pmod{m}$. [1]

(iii) Show that if $ax \equiv ay \pmod{m}$ and $\gcd(a, m) = 1$, then $x \equiv y \pmod{m}$. [2]

(iv) A *reduced residue system modulo m* is a set of integers $\{r_1, \dots, r_k\}$ such that:

- $\gcd(r_i, m) = 1$ for all i where $1 \leq i \leq k$,
- $r_i \not\equiv r_j \pmod{m}$ if $i \neq j$, and
- if x is an integer such that $\gcd(x, m) = 1$, then $x \equiv r_i \pmod{m}$ for some r_i where $1 \leq i \leq k$.

Show that if $\gcd(a, m) = 1$ and S is a reduced residue system modulo m , then $aS = \{ar \mid r \in S\}$ is also a reduced residue system modulo m with the same number of elements as S . [5]

We denote $\phi(m)$ to be the number of positive integers less than or equal to m that are relatively prime to m . For example, $\phi(12) = 4$ since $\gcd(x, 12) = 1$ for $x = 1, 5, 7, 11$.

(v) Find $\phi(16)$. [1]

(vi) By considering the reduced residue system modulo m , $\{r_1, r_2, \dots, r_{\phi(m)}\}$, show that if $\gcd(a, m) = 1$, then

$$a^{\phi(m)} \equiv 1 \pmod{m}. \quad [5]$$

(vii) By using the result in (vi), find the remainder when 3^{101} is divided by 16. [2]